RAJALAKSHMI ENGINEERING COLLEGE

Shalini G V 251001244-2025-IT

S2

Started on	Thursday, 24 September 2026, 10:33 AM
State	Finished
Completed on	Thursday, 24 September 2026, 10:35 AM
Time taken	1 min 27 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
int n=9;
int arr[]={-1,3,4,5,2,2,2,2,3};
int dp[9];
int max_len=0;
for(int i=0;i<n;i++){
dp[i]=1;
for(int j=0;j<i;j++){
if(arr[j]<=arr[i]&&dp[j]+1>dp[i]){
dp[i]=dp[j]+1;
}
}
if(dp[i]>max_len){
max_len=dp[i];
}
}
printf("%d\n",max_len);
return 0;
}
```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course